

Exam History of AI

Date: Thursday July 3rd, 2025	Lecturer: dr. D.J. Beckers
Time: 2h 15min	Course code: FEW XB 0113

In general:

- (1) Make sure your name and student number are on ALL your work.
- (2) Answer in English sentences. Spelling and grammar should be correct either by English or by US standards. Just a few buzzwords never constitute an answer. Always explain yourself. Use appropriate examples to illustrate your answer.
- (3) This exam consists of 8 A-questions, 12 B-questions, 6 C-questions and a NO BONUS question. The A-questions are about the lectures; the B-questions refer to information from the book by Campbell-Kelly. The C-questions require you to combine the information from the book and lectures in your answer.
- (4) You pick four questions: 1 A-question, 2 B-questions and 1 C-question. Each question is worth 2 points. Indicate clearly which questions you're answering by mentioning either the number and / or the title of the question in your answer. If you answer more A-questions, only the first will be considered. Likewise for the B- and C-questions.
- (5) If you're *not* entitled to the bonus point, you additionally do the NO BONUS question. This question is worth one point.

A-Questions:

1. Prehistory of AI

During the lecture, a prehistory of AI was compared to a prehistory of computing. What were the events these histories had in common?

2. Paris conference

A 1951 conference in Paris was in the lecture used as a reason for suggesting that AI was the driving force in computer development in the early years after World War II. Explain the line of thought.

3. Acronyms

Elaborate on the functions of acronyms in 1950s computing.

4. Grace Hopper

In histories of computing, Grace Hopper is often mentioned. Discuss her direct and indirect contributions to both computing and AI.

5. AI and education

To what extent was the 1950s/60s hype of programmed instruction instrumental in the visibility of AI? Make a comparison to the 1970s/80s hype of LOGO.

6. History of AI

Discuss in what ways history of AI helps defining AI.

7. Computers and ordinary life

In the lecture, it was illustrated how computers in the 1980s and 1990s gradually became part of ordinary life by looking at statistics and blockbuster movies. Explain the line of thought.

8. Data and AI

The rise of AI in the early 21st century was intrinsically intertwined with the rise of data collecting as part of surveillance capitalism. Explain the line of thought.

B-Questions:

1. Moore school

"Computing was not performed for its own sake, but always as a means to an end", Campbell-Kelly claims with regard to Moore school. Explain that remark.

2. Computer manufacturers

Campbell-Kelly describes how, in the late 1940s and early 1950s, computer manufacturers met problems with the novel idea to use magnetic tape for storage, instead of the conventional technique of punched cards or punched tape. Elaborate on these problems.

3. IBM 1401

Campbell-Kelly explains the success of the IBM 1401 as the result of good technology and great marketing. Explain his line of thought.

4. Perry Crawford

Campbell-Kelly narrates how Perry Crawford saved project Whirlwind twice. Explain that remark.

5. Software crisis

To what extent may OS/360 be called exemplar to the software crisis and its results?

6. Timesharing

Campbell-Kelly discusses the role of new ways of computing in the 1960s to explain the rise of personal computing later. Explain his line of thought. Discuss at least two of these "new ways" in detail.

7. Whole earth catalogue

To what extent does Campbell-Kelly suggest that *The Whole Earth Catalogue* influenced the rise of personal computing?

8. Broadening the appeal

According to Campbell-Kelly, a number of companies in the 1980s followed the ideas of launching a GUI based operating system for the personal computer. Discuss two of the companies that were not successful and elaborate on what makes it relevant to know about these histories.

9. Internet

Campbell-Kelly discusses how sociologists and psychologists were fascinated by the social issues brought about by the new medium of e-mail. Explain that remark.

10. Off-shoring and out-sourcing

Campbell-Kelly addresses off-shoring and out-sourcing as the two defining characteristics of globalization of the computer industry. Discuss typical IT examples of those phenomena.

11. Netflix

To what extent does Netflix illustrate the rise of Web 2.0? In what sense does Campbell-Kelly relate Netflix to the rise of surveillance capitalism?

12. Computing and governance

In his final chapter, Campbell-Kelly addresses some differences between US and European legislative programs with regard to privacy. Elaborate on these differences by comparing two of these legal programs.

C-Questions:

1. Histories

The traditional history of computing makes use of the concept of appropriation. To what extent does that concept also apply to a history of AI?

2. Data driven culture

To what extent is “data driven culture” a suitable way of describing twenty-first century culture?

3. Prejudice in AI

Histories of computing and histories of AI are dominated by caucasian males. Provide two examples that may help to contradict that prejudice and that help explain how that narrative came into existence.

4. Computer clubs

Since the 1960s several types of computer clubs were founded. Discuss two types of clubs and compare the US and European versions of these clubs. In your essay use the concept of agendas.

5. Defining technology

Discuss two 21st century examples of AI as a defining technology (in Bolter's sense).

6. AI and histories

To what extent are histories used to further agendas?

NO BONUS

Discuss Pamela McCorduck's history of AI in the context of AI winters.